

Student Observer: _____

Period:

PERIOD 0 HONORS

Date: _____

OBSERVER INSTRUCTIONS:

As each classmate performs their live demonstration, record the presenter(s), demo title, and physical category (Rest, Motion, or Direction). In the right column, **explain how you believe the demonstration utilizes inertia to be successful** (explain why the object maintains its state of rest or motion, how the quick action prevents unwanted force transfer, and verify that *no forward force* acts on the mass).

DEMO & PRESENTER	HOW THE DEMONSTRATION UTILIZES INERTIA TO BE SUCCESSFUL
Presenter Name(s), Demo Title & Category	Explain why the mass behaves as observed using Newton's 1st Law (resistance to acceleration, $\Sigma F = 0$, zero forward force)
DEMO 1 Rest Motion Direction Presenter(s): _____ Demo Title: _____	
DEMO 2 Rest Motion Direction Presenter(s): _____ Demo Title: _____	
DEMO 3 Rest Motion Direction Presenter(s): _____ Demo Title: _____	
DEMO 4 Rest Motion Direction Presenter(s): _____ Demo Title: _____	
DEMO 5 Rest Motion Direction Presenter(s): _____ Demo Title: _____	
DEMO 6 Rest Motion Direction Presenter(s): _____ Demo Title: _____	
DEMO 7 Rest Motion Direction Presenter(s): _____ Demo Title: _____	
DEMO 8 Rest Motion Direction Presenter(s): _____ Demo Title: _____	

DEMO & PRESENTER Presenter Name(s), Demo Title & Category	HOW THE DEMONSTRATION UTILIZES INERTIA TO BE SUCCESSFUL Explain why the mass behaves as observed using Newton's 1st Law (resistance to acceleration, $\Sigma F = 0$, zero forward force)
DEMO 9 ☐ Rest ☐ Motion ☐ Direction Presenter(s): Demo Title:	
DEMO 10 ☐ Rest ☐ Motion ☐ Direction Presenter(s): Demo Title:	
DEMO 11 ☐ Rest ☐ Motion ☐ Direction Presenter(s): Demo Title:	
DEMO 12 ☐ Rest ☐ Motion ☐ Direction Presenter(s): Demo Title:	
DEMO 13 ☐ Rest ☐ Motion ☐ Direction Presenter(s): Demo Title:	
DEMO 14 ☐ Rest ☐ Motion ☐ Direction Presenter(s): Demo Title:	
DEMO 15 ☐ Rest ☐ Motion ☐ Direction Presenter(s): Demo Title:	

HONORS SYNTHESIS: BRIDGING INERTIA TO MECHANICAL EQUILIBRIUM ($\Sigma F = 0$)

1. Minimizing Unwanted Force Transfer: Which demonstration observed today achieved the cleanest isolation of inertia? Explain how the presenter minimized contact time (Δt) and friction to prevent unwanted force from disturbing the mass:

2. The Equilibrium Condition: In every demonstration observed today, whenever the mass stayed at rest or moved at constant velocity, what was the horizontal net force acting on that mass? Write the equation: $\Sigma F_x =$ _____ N. What happens the instant an unbalanced net force acts?

Demonstrations Logged: _____ / 15

Honors Synthesis Check: ☐ Complete

Teacher Verification: _____